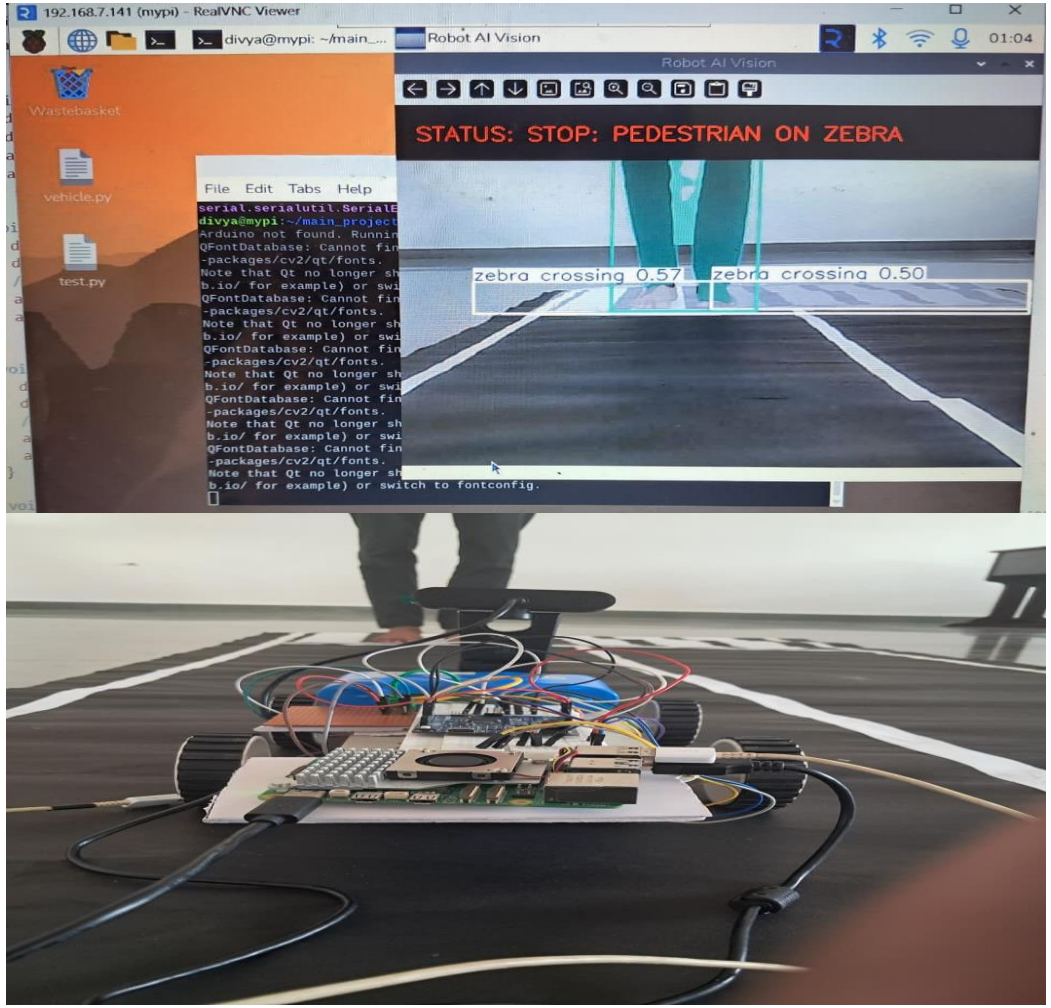


Figure 1: Combined zebra crossing and pedestrian detection using vision-based pattern recognition



Figure 2: Real-time lane detection results highlighting extracted lane boundaries and computed vehicle positions related to the road.

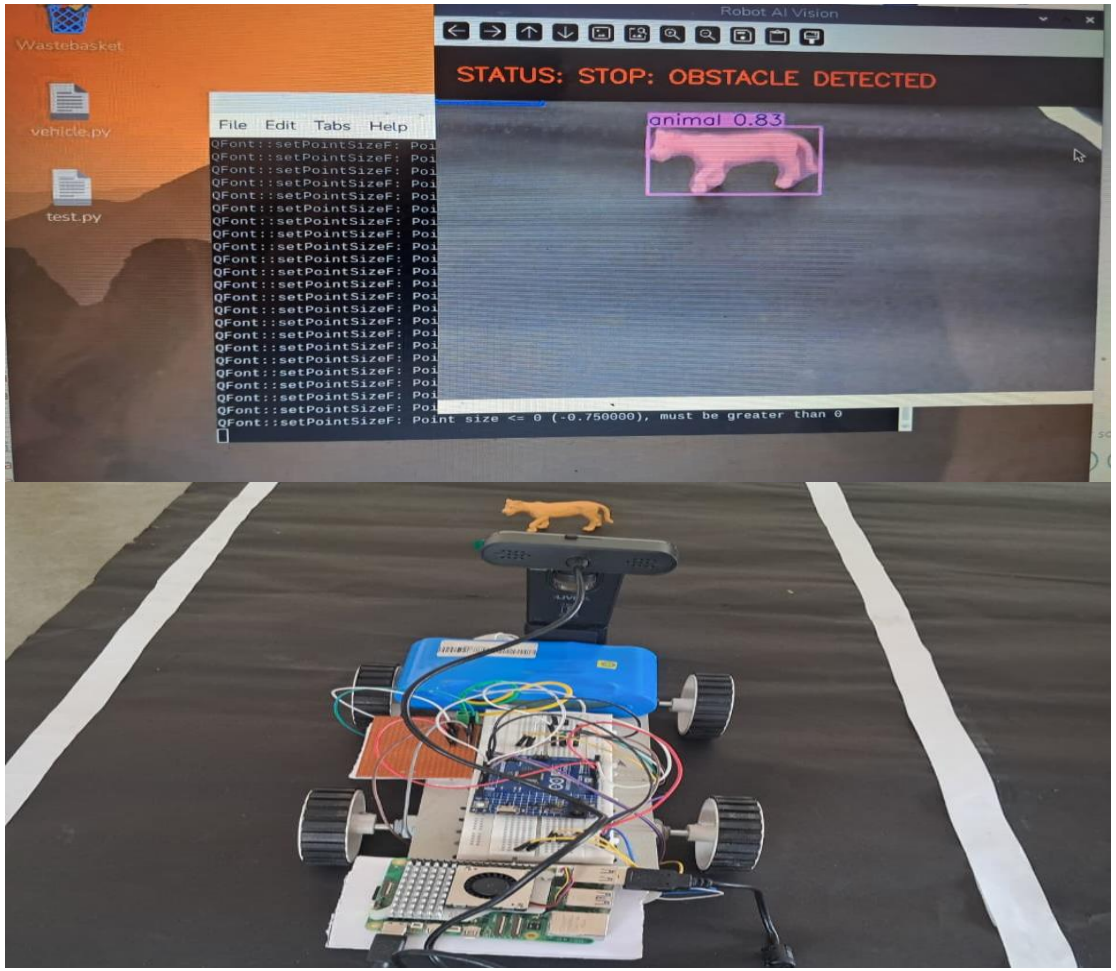


Figure 3:Object detection results in demonstrating recognition of animals in the driving environment

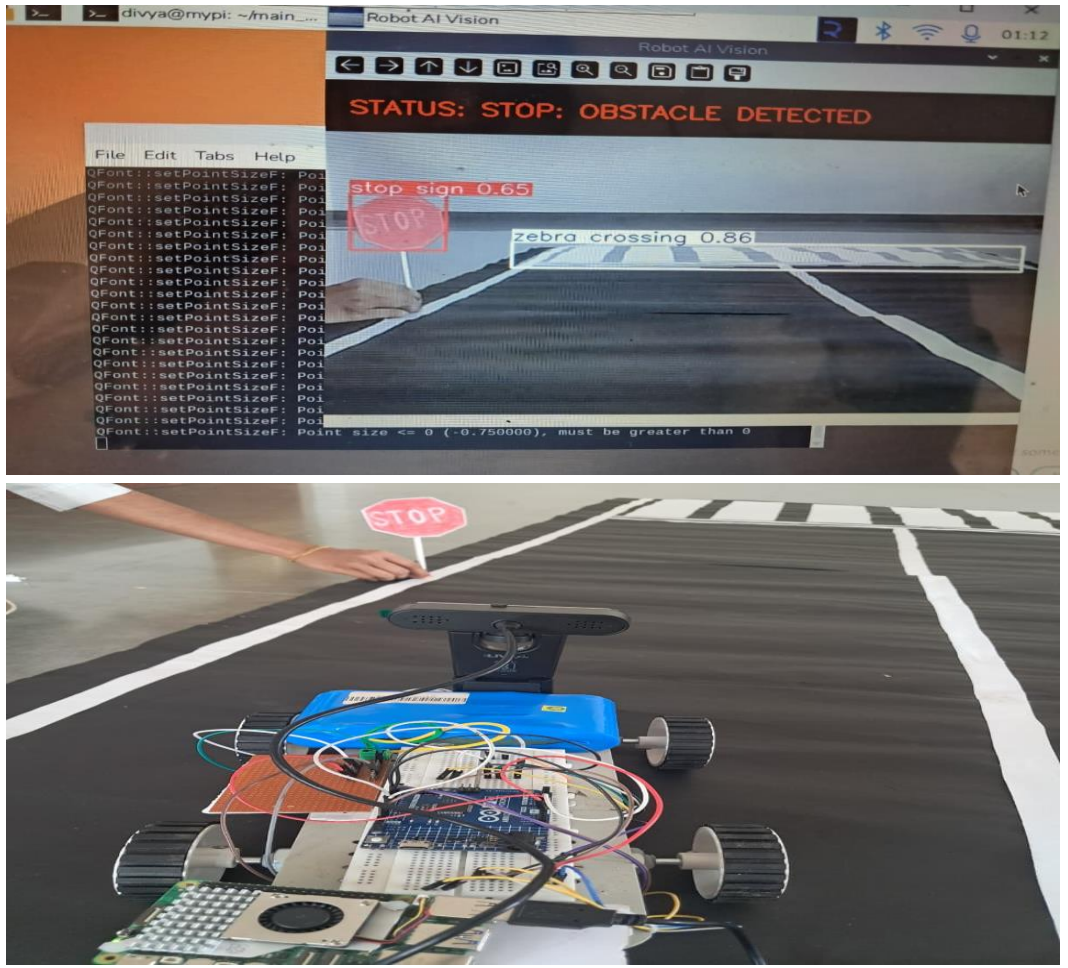


Figure 4: Traffic signboard detection using a YOLO-based object detection model with confidence-based classification.



Figure 5:Object detection results in demonstrating recognition of vehicles in the driving environment

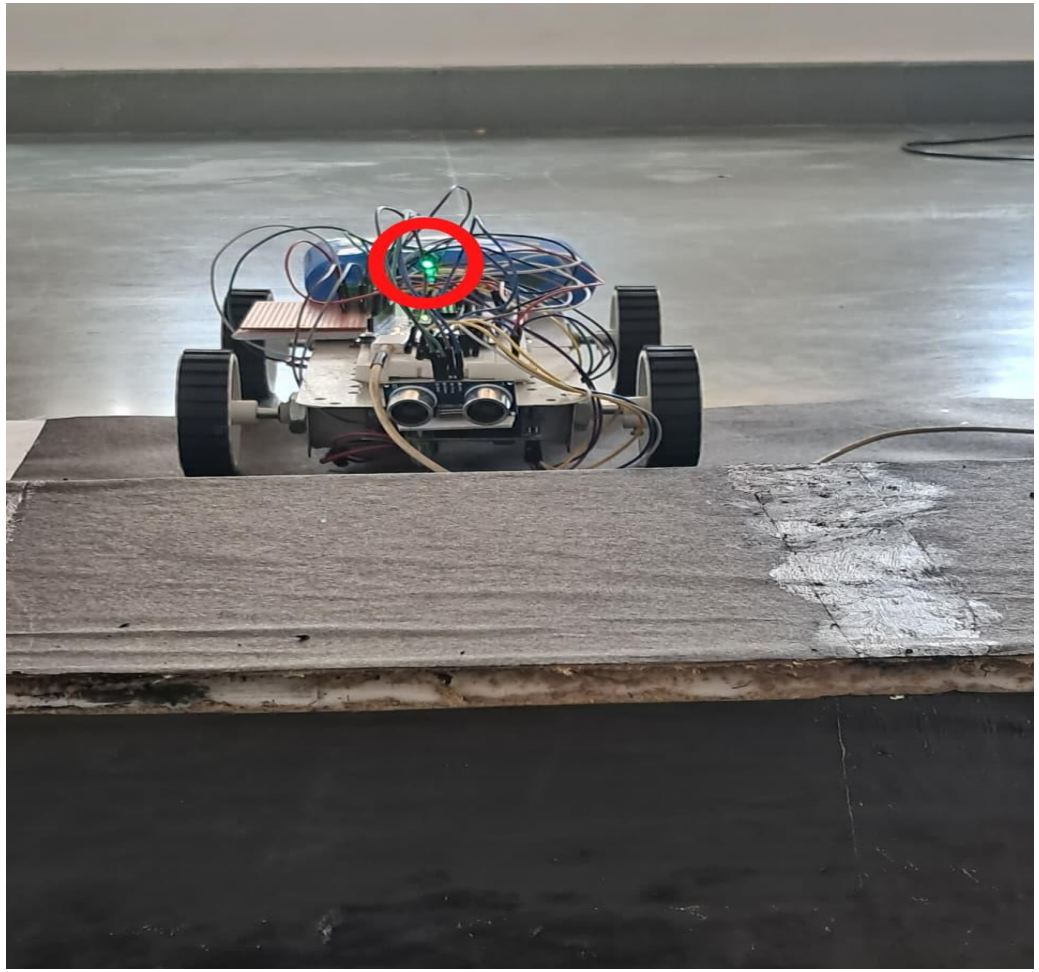


Figure 6: Hump detection using sensor-based detection with LED blinking

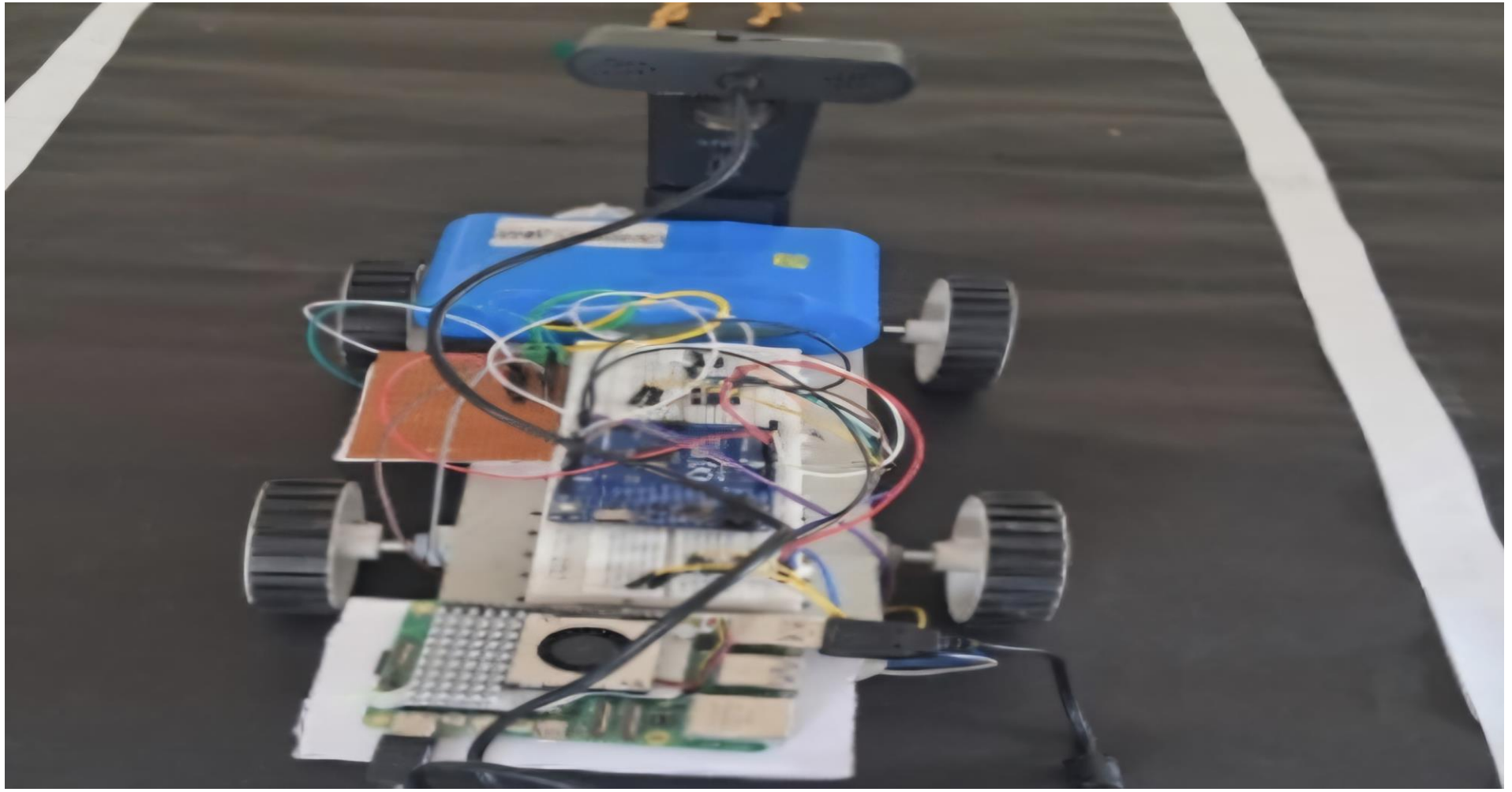


Figure 7: Completed hardware

Table 1: Result and Analysis

Scenario	Accuracy(%)	Notes
Daytime (clear weather)	80%-90%	Animals, vehicles, stop sign are detected with minimal misclassification.
Night time	70%	Reduced visibility causes occasional missed detections.
Rainy/foggy conditions	<60%	Detection struggles due to water droplets on the camera.